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Inventor(s)	Palermo; Thomas J.

Vascular and aortic grafts with telescoping sheath and deployment methods thereof

Abstract

A vascular connector deployment tool may include a handle, an elongated mandrel extending distally from the handle, a vascular connector disposed coaxially about the mandrel, a retractable sheath assembly including an outer sheath telescopically deployed over an inner sheath and an actuator on the handle that is configured to selectively retract the outer sheath and the inner sheath relative to the mandrel. The sheath assembly is configured to constrain the vascular connector against the mandrel in an insertion profile. A distal transition between the outer sheath and the inner sheath visually indicates position of the vascular connector.

Inventors:	**Palermo; Thomas J. (San Jose, CA)**
Applicant:	**Aquedeon Medical, Inc.** (Sunnyvale, CA)
Family ID:	**75223469**
Assignee:	**Aquedeon Medical, Inc. (Sunnyvale, CA)**
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Primary Examiner: Holwerda; Kathleen S

Assistant Examiner: Miller; Serenity A

Attorney, Agent or Firm: Dergosits & Noah LLP

Background/Summary

RELATED APPLICATIONS (1) This patent application claims priority to U.S. Provisional Patent Application No. 62/984,898, filed Mar. 4, 2020, which is hereby incorporated by reference in its entirety and for all purposes.

FIELD OF THE PRESENT DISCLOSURE
(1) The invention generally relates to vascular and aortic connectors, with particular regard to deployment tools and methods for such deploying such connectors.
BACKGROUND
(2) The circulatory system includes the aorta and other large-diameter blood vessels, as well as smaller-diameter blood vessels and capillaries. Therapeutic interventions to replace or support diseased or otherwise compromised vessels may involve the use of synthetic grafts to maintain or restore patency of the affected vessel to perfuse downstream anatomy. Although disease and other conditions are known to affect all types of blood vessels, those affecting the aorta may be more serious and more likely to result in patient death, due to the volume and pressure of blood that is pumped through the aorta. Accordingly, the example discussed below is framed in the context of aortic grafts, but it should be appreciated that the techniques of this disclosure are applicable to other portions of a patient's vasculature.
(3) Aortic aneurysm is a serious condition that can affect any segment of the aorta. An aortic aneurysm in the abdomen is referred to as an abdominal aortic aneurysm or AAA; an aortic aneurysm in the chest cavity is referred to as a thoracic aortic aneurysm or TAA, and an aneurysm in the chest cavity on the aortic arch may be referred to as an aortic arch aneurysm. Aortic aneurysms may result from different causes, such as untreated or severe hypertension, smoking, generic disease such as Marfan's syndrome, and degenerative dilation of the aortic wall. A thoracic aortic aneurysm results from weakening of the aortic wall, leading to localized dilatation, and is a life-threatening condition. Patients with thoracic aneurysms are often asymptomatic until the aneurysm expands. The most common presenting symptoms are pain and aortic rupture. A ruptured aneurysm can cause severe internal bleeding, which can rapidly lead to shock or death.
(4) Patients with acute dissection typically present with pain and are classed as emergencies due to the risk of the dissection rupturing the wall of the aorta, affecting the integrity of the aortic valve and, through involvement of the origins of the coronary arteries, affecting perfusion of the myocardium. Typically, a dissection of the ascending aorta that extends into the aortic arch or that involves any of the arteries of the aortic arch requires surgical insertion of a graft to replace the diseased portion of the ascending aorta and additional grafts to reestablish blood flow to each artery

stemming from the aortic branch where any dissection or disease is present. The grafts may be connected to the vasculature by expanding a connector from an insertion profile to a deployed profile to secure the graft to a vessel. The procedure may also involve inserting an additional stent that is dedicated to provide blood flow from the ascending aorta to vasculature distal of the descending aorta. Conditions affecting other portions of the patient's vasculature may also be treated in a similar manner.

(5) Complex thoracic aortic disease encompasses acute (AAD) and chronic type A dissections (CAD), as well as aortic arch aneurysm with or without involvement of the ascending and descending aorta. Aortic dissection results from a tear in the inner layer of the wall of the aorta leading to blood entering and separating the layers of the wall. Acute aortic dissections are defined as those identified within the first 2 weeks after the initial tear, and chronic dissections are defined as those identified at times greater than 2 weeks. Aortic dissection is classified by its location and the extent of involvement of the thoracic aorta. Stanford Type A dissection affects the ascending aorta and may extend to the arch and descending thoracic aorta. Stanford Type B dissection does not affect the ascending aorta and typically involves the descending thoracic aorta, distal to the origin of the left subclavian artery. Approximately two-thirds of aortic dissections are Stanford Type A.

(6) Treatment of complex thoracic aortic disease typically requires long and complicated open surgery. During such surgery, the patient is typically placed on a cardiopulmonary bypass pump, and the heart is stopped to allow the aorta to be clamped and operated upon. While the patient is on cardiopulmonary bypass, the patient generally is also chilled to a condition of hypothermia. The risk that the patient will not be able to survive the surgery is directly related to the duration of time that the patient spends on pump and under hypothermia.

(7) Correspondingly, it would be desirable to provide tools and methods for deploying aortic connectors for aneurysm repair that facilitate and expedite their placement. Similarly, it would also be desirable to provide tools and methods that may be used for deploying connectors in other portions of a patient's vasculature. As will be detailed in the following materials, this disclosure satisfies these and other goals.

SUMMARY

(8) The present disclosure is directed to a vascular connector deployment tool having a handle, an elongated mandrel extending distally from the handle, a vascular connector disposed coaxially about the mandrel, a retractable sheath assembly including an outer sheath telescopically deployed over an inner sheath, wherein the sheath assembly is configured to constrain the vascular connector against the mandrel in an insertion profile and wherein a distal transition between the outer sheath and the inner sheath visually indicates position of the vascular connector and an actuator on the handle that is configured to selectively retract the outer sheath and the inner sheath relative to the mandrel.

(9) In one aspect, the inner sheath is visible through the outer sheath.

(10) In one aspect, the outer sheath may be transparent.

(11) In one aspect, the outer sheath may cover all of the vascular connector before retraction and the inner sheath may cover a predetermined amount of a proximal portion of the vascular connector before retraction. For example, the inner sheath may cover a proximal half of the vascular connector before retraction.

(12) In one aspect, a distal end of the vascular connector may be visible through the outer sheath.

(13) In one aspect, the actuator may be configured to initially retract only the outer sheath relative to the mandrel to expose a distal portion of the vascular connector and to subsequently retract both the outer sheath and the inner sheath relative to the mandrel to expose all of the vascular connector. The actuator may be a slider. The slider may be configured to initially engage only the outer sheath and to engage both the inner sheath and the outer sheath when reset. Alternatively, the slider may have a range of motion and may be configured to engage only the outer sheath over a first portion

of the range of motion and to engage both the inner sheath and the outer sheath over a remaining portion of the range of motion.
(14) In one aspect, the inner sheath may be transparent and a marker on a distal end of the inner sheath may be visible through the outer sheath. A stationary shoulder tube coaxially disposed over the mandrel and within the outer sheath and the inner sheath may abut a proximal end of the vascular connector. The shoulder tube may be visible through the outer sheath and the inner sheath.
(15) In one aspect, the vascular connector may be a self-expanding connector that is able to maintain radial force at the temperature in a range of 22° C. to 37° C.
(16) This disclosure also includes a method for implanting a vascular connector in a patient. The method may involve providing a vascular connector deployment tool including a handle, an elongated mandrel extending distally from the handle, a vascular connector disposed coaxially about the mandrel, a retractable sheath assembly including an outer sheath telescopically deployed over an inner sheath, wherein the sheath assembly constrains the vascular connector against the mandrel in an insertion profile, and an actuator on the handle that is configured to selectively retract the outer sheath and the inner sheath relative to the mandrel, positioning at least a distal portion of the vascular connector within a first lumen for conducting blood of the patient by visualizing a distal transition between the outer sheath and the inner sheath, manipulating the actuator to cause the outer sheath to retract relative to the mandrel and expose a distal portion of the vascular connector and securing the distal portion of the vascular connector within the first lumen by expansion of the portion of the vascular connector from the insertion profile.
(17) In one aspect, positioning at least a distal portion of the vascular connector within the first lumen of the patient may involve advancing the deployment tool until a distal end of the inner sheath is adjacent an opening to the first lumen of the patient.
(18) In one aspect, positioning at least a distal portion of the vascular connector within the first lumen of the patient may involve positioning one half of the vascular connector within the first lumen.
(19) In one aspect, the method may also involve confirming stationarity of the deployment tool during retraction of the outer sheath by visualizing the inner sheath through the outer sheath.
(20) In one aspect, a second lumen for conducting blood of the patient may be advanced coaxially over the deployment tool until an end of the second lumen is adjacent the opening of the first lumen, the actuator may be manipulated to cause the outer sheath and the inner sheath to retract relative to the mandrel and expose a remaining portion of the vascular connector and the remaining portion of the vascular connector may be secured within the second lumen by expansion of the portion of the vascular connector from the insertion profile. The first lumen may be a blood vessel and the second lumen may be a graft.
(21) In one aspect, the deployment tool may have a shoulder tube that abuts the vascular connector such that the method also includes visualizing the shoulder tube through the outer sheath and the inner sheath to confirm placement of the vascular connector within the second lumen.
(22) In one aspect, the vascular connector may be a self-expanding connector that is able to maintain radial force at the temperature in a range of 22° C. to 37° C.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Further features and advantages will become apparent from the following and more particular description of the preferred embodiments of the disclosure, as illustrated in the accompanying drawings, and in which like referenced characters generally refer to the same parts or elements throughout the views, and in which:
(2) FIG. **1** schematically illustrates a vascular connector used to connect a blood vessel and a graft,

according to an embodiment of the disclosure.
(3) FIG. **2** schematically illustrates a deployment tool for positioning and placing a vascular connector, according to an embodiment of the disclosure.
(4) FIG. **3** schematically illustrates positioning a vascular connector within a blood vessel, according to an embodiment of the disclosure.
(5) FIG. **4** schematically illustrates retracting an outer sheath of the deployment tool to allow expansion of a distal portion of the vascular connector to secure it within the blood vessel, according to an embodiment of the disclosure.
(6) FIG. **5** schematically illustrates advancing a graft over the deployment tool to bring it into proximity with the blood vessel, according to an embodiment of the disclosure.
(7) FIG. **6** schematically illustrates retracting the outer sheath and an inner sheath of the deployment tool to allow expansion of a remaining portion of the vascular connector to secure it within the graft, according to an embodiment of the disclosure.
(8) FIG. **7** schematically illustrates another deployment tool for positioning and placing a vascular connector, according to an embodiment of the disclosure.
DETAILED DESCRIPTION
(9) At the outset, it is to be understood that this disclosure is not limited to particularly exemplified materials, architectures, routines, methods or structures as such may vary. Thus, although a number of such options, similar or equivalent to those described herein, can be used in the practice or embodiments of this disclosure, the preferred materials and methods are described herein.
(10) It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments of this disclosure only and is not intended to be limiting.
(11) The detailed description set forth below in connection with the appended drawings is intended as a description of exemplary embodiments of the present disclosure and is not intended to represent the only exemplary embodiments in which the present disclosure can be practiced. The term "exemplary" used throughout this description means "serving as an example, instance, or illustration," and should not necessarily be construed as preferred or advantageous over other exemplary embodiments. The detailed description includes specific details for the purpose of providing a thorough understanding of the exemplary embodiments of the specification. It will be apparent to those skilled in the art that the exemplary embodiments of the specification may be practiced without these specific details. In some instances, well known structures and devices are shown in block diagram form in order to avoid obscuring the novelty of the exemplary embodiments presented herein.
(12) For purposes of convenience and clarity only, directional terms, such as top, bottom, left, right, up, down, over, above, below, beneath, rear, back, and front, may be used with respect to the accompanying drawings. These and similar directional terms should not be construed to limit the scope of the disclosure in any manner.
(13) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one having ordinary skill in the art to which the disclosure pertains. As used in this document, and as customarily used in the art, the word "substantially" and similar terms of approximation refer to normal variations in the dimensions and other properties of finished goods that result from manufacturing tolerances and other manufacturing imprecisions. Finally, as used in this specification and the appended claims, the singular forms "a, "an" and "the" include plural referents unless the content clearly dictates otherwise.
(14) Referring to FIG. **1**, a vascular connector **10** is shown that may be adapted for use in the aorta or other suitable locations in the patient's vasculature to help establish or restore fluid communication by joining blood vessel **12**, such as one that may be dissected, to graft **14**, such as a branch graft. Graft **14** may be fabricated from any suitable material or materials, such as but not limited to polytetrafluoroethylene (PTFE) or a polyester such as polyethylene terephthalate (PET), sometimes known as DACRON® brand polyester available from E. I. Du Pont De Nemours and

Company of Wilmington, Delaware. The vascular connector **10** is expandable from a first insertion diameter to a second deployed diameter and may have any structure that allows for expansion from a first insertion diameter to a second deployed diameter and that holds vascular connector securely inside vessel **12** and graft **14** in the deployed state. As one example, vascular connector **10** may include a plurality of circumferentially extending hoops **16**, similar in design to a stent. The hoops **16** may be longitudinally spaced apart; and if so, adjacent hoops **16** may be connected by one or more tie bars **18**. Alternately, adjacent hoops **16** are not spaced apart, but instead abut or overlap one another. In such a configuration, such adjacent hoops **16** may be fixed to one another, such as by laser welding. The hoops **16** may be fabricated from metal or other material. Each hoop **16** may have a complex shape in which the hoop **16** is fabricated from a wire, or laser cut from a tube, or otherwise manufactured such that the hoop **16** has a complex shape, such as a zig-zag, repeating Z shape, tortuous curve, or other shape. Such a shape allows the hoop **16** to expand from an insertion diameter to a deployed diameter. The zig-zag pattern of at least one hoop **16** may be continuously curved, or may include straight segments connected by curved segments. In one embodiment, the zig-zag pattern of the hoops **16** may be as set forth in expired U.S. Pat. No. 4,580,568, which is incorporated herein by reference in its entirety. However, at least one hoop **16** may be configured differently.

(15) In one embodiment, different hoops **16** may be fabricated from different materials. For example, at least one hoop **16** may be fabricated from superelastic material, such as nickel-titanium alloy, and at least one other hoop **16** may be fabricated from plastically-deformable material, such as 316L stainless steel. Adjacent hoops **16** may alternate between different materials, such that no hoop **16** is adjacent to a hoop **16** composed of the same material. In other embodiments, several hoops **16** composed of the same material may be grouped together, and at least one hoop **16** composed of a different material may be adjacent to that group. For example, a hoop **16** at an outer end of vascular connector **10** may be composed of stainless steel, and the remaining hoops **16** may be composed of superelastic material such as nickel-titanium alloy. By using hoops **16** fabricated from different materials, the vascular connector **10** takes advantage of the different properties of those different materials. For example, one or more hoops **16** fabricated from superelastic material are useful in self-expanding the vascular connector **10**. One or more additional hoops **16** fabricated from a plastically-deformable material such as 316L stainless steel are useful for maintaining the lumen of vascular connector **10** open, because such material has greater resistance to hoop stress and is not susceptible to a return to a different crystal phase after expansion. In addition, such hoops **16** fabricated to be in structure configured to maintain its radial force at the temperature ranging from 22° C. to 37° C. Although the term “hoop” is used in this document, the hoops **16** need not be perfectly circular as viewed on end, and may have a different shape and curvature as suitable for a particular application. In some embodiments, the hoops **16** are substantially circular as viewed on end.

(16) In one embodiment, the opposing ends of vascular connector **10** each expand to the same or similar diameters in the deployed state. In other embodiments, one end expands to a different diameter in the deployed state than the opposing end to allow joining a vessel **12** and graft **14** that have different diameters. The difference in diameter may be controlled by controlling the diameter of the hoops **16** in the respective ends, by providing a different mix of hoops **16** with different materials, or in any other suitable manner.

(17) As noted above, vascular connector **10** may be used to join vessel **12** to graft **14** and it is correspondingly desirable to first deploy half the length of connector **10** into vessel **12**, and then the other half of connector **10** within graft **14** to effectively connect the vessel **12** to the graft **14**. Accurately deploying the proper amount of connector **10** into both the vessel **12** and graft **14** is advantageous since too much of the connector **10** being deployed in the vessel **12** risks the connection within the graft **14** while too little of the connector **10** being deployed in the vessel **12** risks connection within the vessel **12**, and it should be appreciated that both scenarios can be

catastrophic to the patient. Appropriate visualization greatly facilitates any open surgical procedure, so the techniques of this disclosure improve the ability to accurately assess the proper amount of connector **10** within the vessel **12** or branch graft **14** during the operation as discussed in detail below.

(18) Accordingly, vascular connector **10** or a connector having similar characteristics may be deployed using deployment tool **20** as schematically depicted in FIG. **2**. Deployment tool **20** includes handle **22** having an actuator configured as slider **24** that is coupled to outer sheath **26** and inner sheath **28**. At the distal end of deployment tool **20**, is a blunt, atraumatic tip **30**. As indicated, vascular connector **10** is wrapped around an internal mandrel **32** and compressed at least partially against the mandrel **32** by sheaths **26** and **28**. Outer sheath **26** is telescopically disposed over inner sheath **28** and both are selectively retractable via actuation of slider **24**. In one embodiment, a first actuation of slider **24** retracts only outer sheath **26** and a second actuation of slider **24** then retracts both outer sheath **26** and inner sheath **28** simultaneously as described in further detail below. Before retraction, outer sheath **26** covers the entire length of vascular connector **10** while inner sheath **28** covers a predetermined amount of the proximal portion of connector **10** determined based on the characteristics of the vascular connector **10** and the objectives of the procedure. As one example, one half of the overall connector length is covered by inner sheath **28** in the embodiment shown. To illustrate, inner sheath **28** may extend 3.5 cm over the proximal end of a 7 cm connector. Thus, the transition of inner sheath **28** and outer sheath **26** can be used to indicate the position of the vascular connector **10** to be deployed in the vessel **12** or the graft **14**. The distances can be varied as warranted depending on the length of the connector **10** and the proportion desired to be covered. Sheaths **26** and **28** may be formed from suitable polymeric materials, such as nylon (polyamide), urethane, polypropylene, as well as polyamide co-polymers such as, for example, polyether block amides (PEBAX®), or others may be employed. Notably, outer sheath **26** is sufficiently translucent to allow at least the distal end of inner sheath **28** to be visualized through it (as indicated by the dashed lines), for example by using a substantially clear or transparent polymer. It should be recognized that clarity is relative and for the purposes of this disclosure, any difference in optical properties that enables inner sheath **28** to be visualized through outer sheath **26** may be used. As desired, inner sheath **28** may also employ any suitable optical property to facilitate visualization through outer sheath **26**. For example, inner sheath **28** may have added coloration, such as a high visibility or high contrast characteristic. Alternatively, or in addition, inner sheath **28** may have relatively greater opacity than outer sheath **26**. Further, outer sheath **26** may also allow visualization of connector **10**, for example the distal portion not covered by inner sheath **28**.

(19) As will be appreciated, these characteristics of deployment tool **20** facilitate accurate placement of connector **10** within vessel **12** and graft **14**. To the operator, the transition between the translucent outer sheath **26** and the visually distinguishable inner sheath **28** act as marker for a known location on connector **10**. For example, in the shown embodiment in which inner sheath **28** covers one half of connector **10** before retraction provides identification of the mid-point of connector **10**. Accordingly, this helps ensure that one half of the connector **10** is positioned within vessel **12**, leaving the other half to secure graft **14**. If different relative amounts of connector **10** are desired to be allocated to vessel **12** and graft **14** respectively, inner sheath **28** can be configured to cover the appropriate, predetermined amount of the connector **10**. Furthermore, when deploying the distal portion of connector **10** by retracting outer sheath **26** alone, inner sheath **28** is visible and therefore allows continual confirmation that it remains stationary to indicate that the delivery system has not moved relative to the target landing area.

(20) As an illustration only and without limitation, FIGS. **3-6** schematically depict use of deployment tool **20** to connect vessel **12** to graft **14** with connector **10**. As indicated in FIG. **3**, deployment tool **20** may be advanced into vessel **12** until the distal end of inner sheath **28** is adjacent the opening of vessel **12**. The distal end of inner sheath **28** can be visualized through the outer sheath **26** so that the vascular connector **10** can be easy to be positioned accurately in the

vessel **12**. Actuating slider **24** of deployment tool **20** causes retraction of outer sheath **26** alone, leaving inner sheath **28** stationary with respect to handle **22**. As shown in FIG. **4**, retraction of outer sheath **26** allows the distal portion of connector **10** that has now been exposed to expand from its insertion profile to its deployed profile to engage and secure vessel **12**. Next, as indicated in FIG. **5**, graft **14** may be advanced coaxially over deployment tool **20** until it is adjacent vessel **12**, such as by abutting it or otherwise being in sufficient proximity. Now, subsequent actuation of slider **24** causes simultaneous retraction of both outer sheath **26** and inner sheath **28** to expose the remaining proximal portion of connector **10** so that it can expand from its insertion profile to its deployed profile to engage and secure graft **14**. Deployment tool **20** can now be withdrawn proximally, leaving connector **12** in place to connect vessel **12** and graft **14**. Any suitable mechanical implementation may be used to cause slider **24** to selectively retract outer sheath **26** or retract both outer sheath **26** and inner sheath **28**. For example, slider **24** may engage only outer sheath **26** in an initial configuration. Resetting slider **24** after a first actuation may then engage inner sheath **28**, so that subsequent actuation retracts both simultaneously. As another example, slider **24** may engage only outer sheath **26** over a first portion of its range of motion and then engage both outer sheath **26** and inner sheath **28** over the remainder of its range of motion. One of ordinary skill in the art will recognize that other techniques and mechanisms can be used to provide similar results. Further, the embodiment shown in FIGS. **3-6** is provided in the context of joining vessel **12** and graft **14** in an end-to-end configuration, however the techniques can also be applied to join any combination of vessels, grafts or other lumens that conduct the patient's blood and may also be used with an intra-vascular approach through an opening formed in a sidewall of a vessel, graft or other lumen.

(21) Another embodiment of the techniques of this disclosure is schematically depicted in FIG. **7** in the context of deployment tool **40**. Similar elements having similar functionality are indicated using the same reference numbers. As with the previous embodiment, deployment tool **40** also includes handle **22** having an actuator configured as slider **24** that is coupled to outer sheath **26** and inner sheath **28**. At the distal end of deployment tool **20**, is a blunt, atraumatic tip **30**. As indicated, vascular connector **10** is wrapped around an internal mandrel **32** and compressed at least partially against the mandrel **32** by sheaths **26** and **28**. Outer sheath **26** is telescopically disposed over inner sheath **28** and both are selectively retractable via actuation of slider **24**. Again, a first actuation of slider **24** retracts only outer sheath **26** and a second actuation of slider **24** then retracts both outer sheath **26** and inner sheath **28** simultaneously. Also similarly, outer sheath **26** covers the entire length of vascular connector **10** before retraction and inner sheath **28** covers the desired predetermined amount of the proximal portion of connector **10** as discussed above. In this embodiment, both outer sheath **26** and inner sheath **28** are clear or otherwise sufficiently transparent and the transition between the outer sheath **26** and inner sheath **28** is visually indicated by circumferential ring **42** or any other suitable marker that can be visualized through outer sheath **26**. Deployment tool **40** also features a stationary shoulder tube **44** that abuts the proximal end of connector **10** and which is colored or otherwise visible through both outer sheath **26** and inner sheath **28**. Shoulder tube **44** functions to resist proximal movement of connector **10** during retraction of outer sheath **26** and inner sheath **28**. Further, the visibility of shoulder tube **44** also functions as a ready visual indicator of the location of the proximal end of vascular connector **10**. During use, this indication may be used to confirm that the proximal portion of connector **10** is wholly within graft **14** for example and does not extend out of the opening through which deployment tool **40** is advanced. As will be appreciated, deployment tool **40** may be used to position and deliver connector **10** to connect blood carrying lumens of a patient as described above.

(22) While the invention has been described in detail, it will be apparent to one skilled in the art that various changes and modifications can be made and equivalents employed, without departing from the present invention. It is to be understood that the invention is not limited to the details of construction, the arrangements of components, and/or the method set forth in the above description or illustrated in the drawings. Statements in the abstract of this document, and any summary

statements in this document, are merely exemplary; they are not, and cannot be interpreted as, limiting the scope of the claims. Further, the figures are merely exemplary and not limiting. Topical headings and subheadings are for the convenience of the reader only. They should not and cannot be construed to have any substantive significance, meaning or interpretation, and should not and cannot be deemed to indicate that all of the information relating to any particular topic is to be found under or limited to any particular heading or subheading. Therefore, the invention is not to be restricted or limited except in accordance with the claims and their legal equivalents.

Claims

1. A vascular connector deployment tool, comprising: a handle; an elongated mandrel extending distally from the handle, the mandrel having a proximal end coupled to the handle, a distal end and an outer surface; a vascular connector having a proximal circumferential portion and a distal circumferential portion disposed over a portion of the mandrel distal end, the vascular connector being expandable from a first insertion diameter to a second deployed diameter and has a structure that allows for expansion from the first insertion diameter to the second deployed diameter, the structure being configured to hold the vascular connector securely inside a blood vessel and a graft in the deployed state; a retractable sheath assembly including a translucent retractable outer sheath telescopically deployed over a retractable inner sheath that is visible through the outer sheath, wherein the inner and outer sheaths of the retractable sheath assembly are configured to constrain the vascular connector against the outer surface of the mandrel in an insertion profile and wherein a distal transition between the outer sheath and the inner sheath visually indicates position of the vascular connector; and a single actuator on the handle, the single actuator being configured as a slider having a longitudinal range of motion in a slot on the handle, the slider being mechanically coupled to both the outer sheath and the inner sheath to initially retract only the outer sheath relative to the mandrel to expose the distal circumferential portion of the vascular connector and cause the distal circumferential portion to expand from the first insertion diameter to the second deployed diameter, and to subsequently retract the outer sheath and the inner sheath relative to the mandrel to expose the proximal circumferential portion of the vascular connector and cause the proximal circumferential portion to expand from the first insertion diameter to the second deployed diameter.
2. The vascular connector deployment tool of claim 1, wherein the outer sheath covers all of the vascular connector before retraction, and wherein the inner sheath covers a predetermined amount of the proximal circumferential portion of the vascular connector before retraction.
3. The vascular connector deployment tool of claim 2, wherein the inner sheath covers a proximal half of the vascular connector before retraction.
4. The vascular connector deployment tool of claim 3, wherein a distal end of the vascular connector is visible through the outer sheath.
5. The vascular connector deployment tool of claim 1, wherein, upon a first actuation of the slider, the slider is configured to initially engage only the outer sheath and when the slider is reset after the first actuation, the slider then engages the inner sheath such that subsequent actuation retracts both the outer sheath and inner sheath simultaneously.
6. The vascular connector deployment tool of claim 1, wherein the slider has a range of motion and is configured to engage only the outer sheath over a first portion of the range of motion and to engage both the inner sheath and the outer sheath over a remaining portion of the range of motion.
7. The vascular connector deployment tool of claim 1, wherein the inner sheath is transparent and a marker on a distal end of the inner sheath is visible through the outer sheath.
8. The vascular connector deployment tool of claim 7, further comprising a stationary shoulder tube coaxially disposed over the mandrel and within the outer sheath and the inner sheath that abuts a proximal end of the vascular connector.

9. The vascular connector deployment tool of claim 8, wherein the shoulder tube is visible through the outer sheath and the inner sheath.
10. The vascular connector deployment tool of claim 1, wherein the vascular connector is self-expanding connector and is able to maintain radial force at the temperature in a range of 22° C. to 37° C.
11. A method for implanting a vascular connector in a patient, comprising: providing a vascular connector deployment tool including a handle, an elongated mandrel extending distally from the handle, the mandrel having a proximal end coupled to the handle, a distal end and an outer surface having a circumference, a vascular connector having a proximal circumferential portion and a distal circumferential portion disposed coaxially around the circumference of the mandrel outer surface at the distal end, the vascular connector being expandable from a first insertion diameter to a second deployed diameter and has a structure that allows for expansion from the first insertion diameter to the second deployed diameter, the structure being configured to hold the vascular connector securely inside a first lumen and a second lumen in the deployed state, a retractable sheath assembly including a translucent retractable outer sheath telescopically deployed over a retractable inner sheath that is visible through the outer sheath, wherein the retractable sheath assembly constrains the vascular connector against the outer surface of the mandrel in an insertion profile, and a single actuator on the handle, the single actuator being configured as a slider having a longitudinal range of motion in a slot on the handle, the slider being mechanically coupled to both the outer sheath and the inner sheath to initially retract only the outer sheath relative to the mandrel to expose the distal circumferential portion of the vascular connector and cause the distal circumferential portion to expand from the first insertion diameter to the second deployed diameter, and to subsequently retract the outer sheath and the inner sheath relative to the mandrel to expose the proximal circumferential portion of the vascular connector and cause the proximal circumferential portion to expand from the first insertion diameter to the second deployed diameter; positioning at least the distal circumferential portion of the vascular connector within the first lumen for conducting blood of the patient by visualizing a distal transition between the outer sheath and the inner sheath; manipulating the actuator to cause the outer sheath to retract relative to the mandrel and expose the circumferential distal portion of the vascular connector; and securing the distal portion of the vascular connector within the first lumen by expansion of the circumferential distal portion of the vascular connector from the insertion profile.
12. The method of claim 11, wherein positioning at least the distal portion of the vascular connector within the first lumen of the patient comprises advancing the deployment tool until a distal end of the inner sheath is adjacent an opening to the first lumen of the patient.
13. The method of claim 11, wherein positioning at least the distal portion of the vascular connector within the first lumen of the patient comprises positioning one half of the vascular connector within the first lumen.
14. The method of claim 11, further comprising confirming the deployment tool remains stationary during retraction of the outer sheath by visualizing the inner sheath through the outer sheath.
15. The method of claim 11, further comprising: advancing the second lumen for conducting blood of the patient coaxially over the deployment tool until an end of the second lumen is adjacent the opening of the first lumen; manipulating the actuator to cause the outer sheath and the inner sheath to retract relative to the mandrel and expose the proximal circumferential portion of the vascular connector; and securing the proximal circumferential portion of the vascular connector within the second lumen by expansion of the proximal circumferential portion of the vascular connector from the insertion profile.
16. The method of claim 15, wherein the first lumen is a blood vessel and the second lumen is a graft.
17. The method of claim 15, wherein the deployment tool has a shoulder tube that abuts the vascular connector and further comprising visualizing the shoulder tube through the outer sheath

and the inner sheath to confirm placement of the vascular connector within the second lumen.
18. The method of claim 11, wherein the vascular connector is self-expanding connector and is able to maintain radial force at the temperature in a range of 22° C. to 37° C.